

How it works

This project uses the **Tkinter** module to display the button grid. **Tkinter's** **mainloop()** function listens for button presses and handles them with a special kind of function, called a **lambda** function, that reveals a symbol. If an unmatched symbol has already been revealed, the program checks to see if the second one matches. The project stores the buttons in a dictionary and the symbols in a list.

▽ Matchmaker flowchart

After shuffling the symbols and creating the grid, the program spends its time listening for button presses. It ends when all the matching pairs have been found.

EXPERT TIPS

Lambda functions

Like **def**, the keyword **lambda** is used to define functions. **Lambda** functions are all written on one line and can be used anywhere you need a function. For example, the function **lambda x: x*2** doubles a number. You can assign it to a variable, such as **double = lambda x: x*2**. Then you call it using **double(x)**, where **x** is a number. So **double(2)** would return **4**. **Lambda** functions are very useful in GUI programming, where several buttons may need to call the same function using different parameters. Without the **lambda** functions in Matchmaker, you would have to create a different function for each button – that's 24 functions!

